Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Rice, Eric E. Ph.D.		
Project Title:	Magnetic Fluid Management (MFM)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	608-827-5000
Organization Name:	Orbital Technologies Corporation		
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City:	Madison	**State:**	WI
Zip Code:	53717	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	07/25/2000	**End Date:**	07/25/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hochstein, John (University of Memphis) Chiaverini, Martin (ORBITEC) Gustafson, Robert (ORBITEC) Gramer, Daniel (ORBITEC)		
Grant/Contract No.:	NAS 3 00153		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The top-level approach involves: (1) Computational Fluid Dynamics (CFD) modeling, (2) conduct of ground-based experiments that use ferrofluid and liquid oxygen, (3) development of reduced gravity aircraft experiment hardware, and (4) conduct of experiments on NASA's reduced gravity aircraft facility. The difficult problem of cryogenic fluid handling in many aerospace applications in low gravity environments can be solved by employing a new magnetic fluid technology called, Magnetic Fluid Management (MFM) that has been proven feasible, through analysis and limited testing. This innovative MFM technology has the potential to provide significant advancements over related technologies such as screens, vanes, porous plugs, and no vent fill processes. MFM technology utilizes the magnetic properties of cryogenic fluids for phase separation, consequently enabling new processes which greatly simplify many critical tasks now encountered such as gas-free liquid transfer between cryogenic containers, liquid-free gas venting during storage, liquid-free gas venting during tank refill, and use of autogenous pressurization and control system to maintain tank pressure during liquid expulsion. Demonstrating these important fluid physics aspects is the focus of this proposed NASA Research Announcement (NRA) program. More specifically, this NRA effort will develop and conduct low-g flight experiments demonstrating paramagnetic dewar systems designed for extended life and capable of two-phase cryogenic fluid management in low-gravity. Orbital Technologies Corporation (ORBITEC) has defined the key technical objectives for this proposed effort as: • Enhance the CMFD (Modified-RIPPLE) software capability to include fluid magnetic surface tension effects and user friendliness. • Develop highly accurate CMFD software simulations of the primary technology applications. • Define requirements and analyze, engineer, design, construct, and ground test a safe low-g parabolic flight test-bed system. • Conduct and test low-g parabolic flight testing. • Analyze parabolic flight performance data. • Develop requirements and design for a spaceflight experiment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)